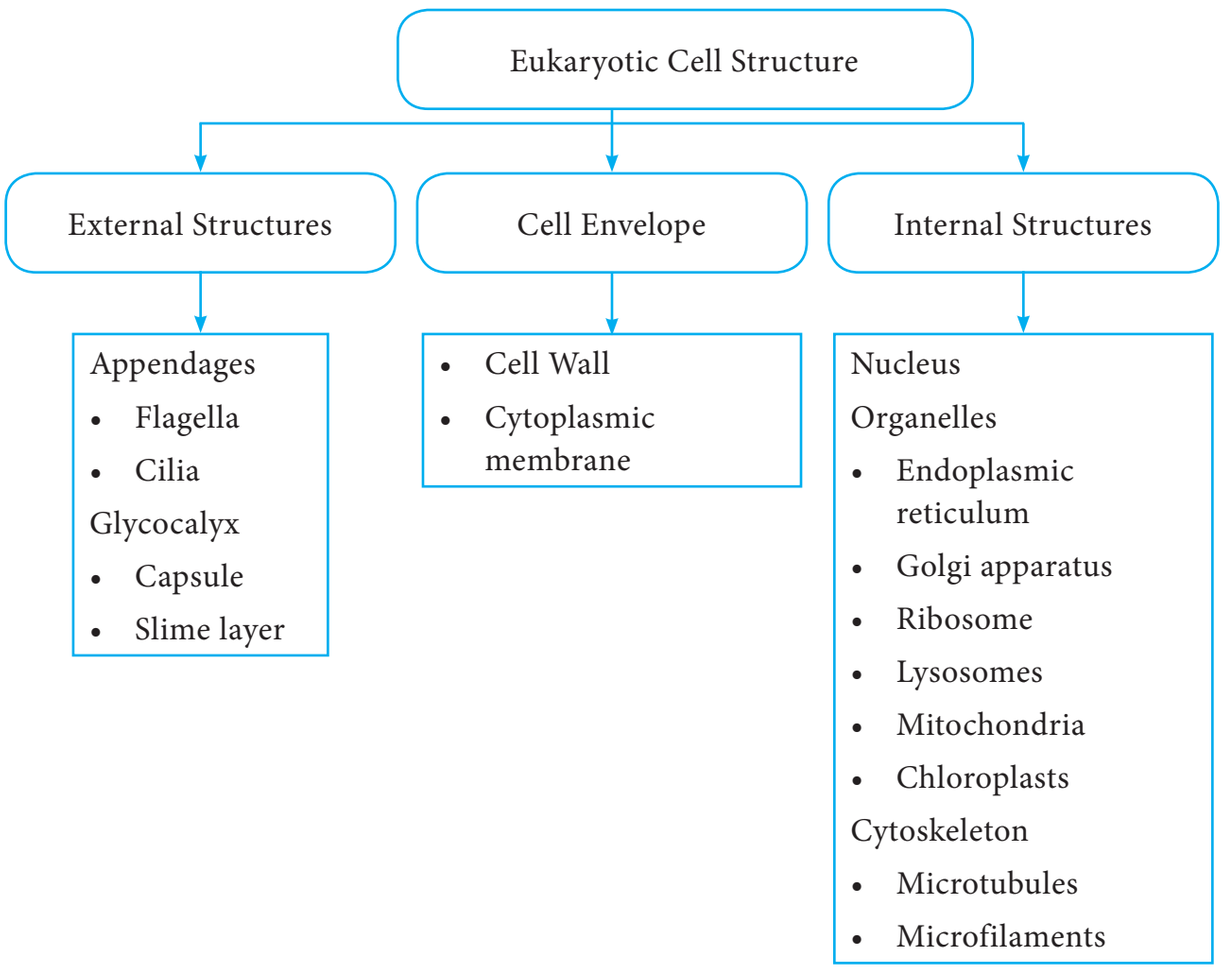
7.5 Eukaryotic Cell Structure

As mentioned earlier, eukaryotic organisms include algae, protozoa, fungi, higher plants and animals. The eukaryotic cell is typically larger and structurally more complex than the prokaryotic cell (Flowchart 7.1).

Prokaryotes and Eukaryotes are chemically similar, in the sense that they both contain nucleic acids, proteins, lipids, and carbohydrates (Figure 7.15). They use the same kinds of chemical reactions to metabolize food, build proteins, and store energy. It is primarily the structure of cell walls and membranes, and the absence of organelles (specialized cellular structures that have specific functions), that distinguish prokaryotes from eukaryotes (Table 7.6).

The general, eukaryotic microbial cells have a cytoplasmic membrane, nucleus, mitochondria, endoplasmic reticulum, Golgi apparatus, vacuoles, cytoskeleton, and glycocalyx. A cell wall, locomotor appendages and chloroplasts are found only in some groups. The structure and functions of the eukaryotic cells are discussed in (Table 7.7).



Flowchart 7.1: Eukaryotic Cell Structure

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